

Ioannis Markakis

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EXPERIENCE

- **Research Associate** September 2025 - September 2027
University of Cambridge Cambridge, UK
 - Project: “Latent Space Semantics for Dependent Type Theory”
 - Lectured the Part II course “Denotational Semantics” and prepared the exam
- **Research Assistant** October 2024 - August 2025
University of Cambridge Cambridge, UK
 - Project: “New Categorical Foundations for Compositional Computer Science”
- **Supervisor** October 2020 - February 2026
University of Cambridge Cambridge, UK
 - Supervised undergraduate computer science students in small groups for the courses “Discrete Mathematics” (Michaelmas 2020, 2021, and 2024), “Logic and Proof” (Lent 2022, 2023, 2024, 2025 and 2026), and “Computation Theory” (Lent 2023)
 - Organized lab sessions for the graduate course “Advanced Topics in Category Theory” (Lent 2023, 2025, 2026)
- **Graduate Teaching Assistant** September 2017 - May 2020
University of Maryland College Park, Maryland, USA
 - Independent instruction for the course “College Algebra and Trigonometry” (Fall 2017)
 - Teaching assistance for the courses “Precalculus” (Fall 2018 and 2020) and “Calculus I” (Spring 2018)
 - Graded weekly assignments for the undergraduate courses “Introduction to Abstract Algebra” (Spring 2019) and “Axiomatic Set Theory” (Spring 2019) as well as the graduate courses “Logic I” (Fall 2020) and “Algebraic Geometry I” (Fall 2020)
- **Undergraduate Teaching Assistant** Spring 2014 and 2016
University of Crete Heraklion, Crete, Greece
 - Exercise tutorials for groups of 50 - 100 students for the course “Foundations of Mathematics”

EDUCATION

- **PhD in Computer Science** October 2020 - July 2025
University of Cambridge Cambridge, UK
 - Thesis title: “An inductive approach to ω -categories and their computads”
 - Supervisor: Prof. Jamie Vicary
 - Submission: September 2024
- **MA in Mathematics** September 2017 - May 2020
University of Maryland College Park, Maryland, USA
 - GPA: 4.0/4.0
- **Ptychion in Mathematics (Bachelor’s degree)** September 2013 - May 2017
University of Crete Heraklion, Greece
 - GPA: 9.34/10 “Excellent”
 - Class Rank: 1st out of 97 students in the academic year 2016-2017
 - Erasmus exchange at the University of Barcelona during the 3rd year of studies
 - Completed 6 graduate-level courses

PUBLICATIONS

• Articles

- [1] T. Benjamin and I. Markakis. *Hom ω -categories of a computad are free*. *Journal of the London Mathematical Society* 112.6 (2025).
- [2] T. Benjamin, I. Markakis, W. Offord, C. Sarti, and J. Vicary. *Naturality for higher-dimensional path types*. *Proceedings of the 40th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2025)*. 2025.
- [3] T. Benjamin, I. Markakis, and C. Sarti. *CaTT contexts are finite computads*. *Electronic Notes in Theoretical Informatics and Computer Science*. Vol. 4 - Proceedings of MFPS XL. 2024, 5.
- [4] C. J. Dean, E. Finster, I. Markakis, D. Reutter, and J. Vicary. *Computads for weak ω -categories as an inductive type*. *Advances in Mathematics* 450 (2024).
- [5] I. Markakis. *Computads for generalised signatures*. *Journal of Pure and Applied Algebra* 228.9 (2024).

• Preprints

- [6] T. Benjamin, C. Champin, and I. Markakis. *A type theory for invertibility in weak ω -categories*. 2026.
- [7] T. Benjamin, C. Champin, and I. Markakis. *Computads with invertible generators for weak ω -categories*. 2026.
- [8] T. Benjamin, I. Markakis, W. Offord, C. Sarti, and J. Vicary. *Higher Eckmann-Hilton Arguments in Type Theory*. 2025.
- [9] T. Benjamin and I. Markakis. *Invertible cells in ω -categories*. 2024.
- [10] D. I. Dais and I. Markakis. *Computing minimal generating systems for some special toric ideals*. 2017.

• Formalisation

- [11] I. Markakis. *Agda formalisation of computads for ω -categories*. 2023.

SCIENTIFIC TALKS

- **Higher Structures in Category Theory, Homotopy Theory and Type Theory (invited)** September 2026
Title: *Invertibility in higher categories* Aberdeen, UK
- **Queer and Trans People in Category Theory 2026 (invited)** August 2026
Title: *Invertibility in higher categories* Hamburg, Germany
- **TYPES 2026** May 2026
Title: *Computads with invertible generators* Gothenburg, Sweden
- **Higher Categories, Polygraphs and Homotopy Seminar at IRIF (invited)** April 2026
Title: *Computads with invertible generators* Paris, France
- **Category Theory 2024** April 2024
Title: *Opposite and hom weak ω -categories* Santiago, Spain
- **Masaryk University Algebra Seminar (invited)** April 2024
Title: *A novel approach to ω -categories* Brno, Czech Republic
- **107th Peripatetic Seminar on Sheaves and Logic** April 2023
Title: *Computads for generalised signatures* Athens, Greece
- **30th Yorkshire and Midlands Category Theory Seminar** March 2023
Title: *Computads for generalised signatures* Birmingham, UK
- **Topology, Algebra and Categories in Logic 2022** June 2022
Title: *Computads for weak ω -categories* Coimbra, Portugal

AWARDS AND SCHOLARSHIPS

- **College Scholarships** *March and November 2022*
Cambridge, UK
Fitzwilliam College
 - Two scholarships for academic excellence in doctoral studies
- **Departmental Scholarship** *Sep 2020*
Cambridge, UK
University of Cambridge
 - Four-year doctoral scholarship from the Department of Computer Science and Technology at the University of Cambridge
- **Graduate Scholarship** *October 2020 - September 2024*
Vaduz, Liechtenstein
Onassis Foundation
 - Four-year doctoral scholarship for studies at the University of Cambridge
- **“Stylianos Pichorides” Award** *November 2016*
Heraklion, Greece
University of Crete
 - Award from the Department of Mathematics and Applied Mathematics for excellence in undergraduate studies
- **“Chrysanthos and Anastasia Karydis” Award** *July 2016*
Heraklion, Greece
University of Crete
 - Award for the highest entrance score in the Department of Mathematics and Applied Mathematics

OTHER ACADEMIC ACTIVITIES

- **Logic in Computer Science 2026** *March 2026*
Lisbon, Portugal
Reviewer
 - Reviewed papers for the conference
- **Logic and Semantics Seminar** *February 2025 -*
Cambridge, UK
Organiser
 - Organising the CLASH group’s seminar series and inviting speakers
- **Mathematical Structures in Computer Science** *Janary 2026*
Reviewer
 - Reviewed papers for the journal
- **Theory and Applications of Categories** *June 2025*
Reviewer
 - Reviewed papers for the journal
- **12th Symposium on Compositional Structures** *April 2024*
Birmingham, UK
Program Committee Member
 - Reviewed two papers for the conference
- **Logical Methods in Computer Science** *October 2023*
Reviewer
 - Reviewed papers for the journal
- **28th and 32nd Yorkshire and Midlands Category Theory Seminar** *September 2022 and August 2023*
Cambridge, UK
Workshop Organizer
 - Responsible for local organization
- **Applied Category Theory 2021** *June 2021*
Cambridge, UK
Conference Organizer
 - Responsible for program organization and communication with speakers
 - Support for remote participation